

Project Summary and Load Schedule

Prairie Horizon Data Campus is a six-hall hyperscale data center campus proposed in Fayette County, Ohio. The ultimate campus build-out is 180 MW at the 138 kV point of interconnection, with approximately 120 MW of critical IT load and 60 MW of supporting mechanical, cooling, lighting, security, and administration load.

The campus will be served from a new customer breaker-and-a-half bay at Jefferson 138 kV Station. Customer-owned high-side equipment includes the 138 kV line terminal, motor-operated disconnects, metering interface equipment, relay panels, and three main power transformers. Downstream campus distribution is arranged at 13.8 kV in a main-tie-main configuration using Siemens NXAIR M 15 kV-class metal-clad switchgear.

Critical load continuity is provided through a hall-based emergency system architecture consisting of 60 Rolls-Royce mtu 16V4000 DS2500 generators, 2,500 kWe each at 13.8 kV, and 216 Vertiv Liebert EXL S1 1,200 kW UPS modules at 480 V in a 2N architecture. Each data hall uses dedicated A and B distribution paths.

Campus Phasing Summary

Phase	Gross campus MW	Critical IT MW	Description
Phase 1	60	40	Two data halls online
Phase 2	120	80	Four data halls online
Phase 3	180	120	Six data halls online

Campus Load Breakdown

Load category	MW at Phase 1	MW at Phase 2	MW at Phase 3
Critical IT load	40	80	120
Cooling plant / mechanical	15	28	41
Electrical losses / UPS auxiliaries	3	6	9
Administration / site support	2	6	10
Total coincident demand at POI	60	120	180

Electrical and Operating Characteristics

Characteristic	Design basis
Power factor target at POI	0.98 lagging or better under normal operation
Reactive compensation	Six switched capacitor-bank steps rated 6 Mvar each
Emergency generation	Emergency-use only; no normal export across the POI
Harmonic design basis	UPS and power conversion equipment to support IEEE 519 compliance at the point of common coupling
Load ramping	Hall load introduced in controlled commissioning blocks; no uncontrolled step-load additions expected

Major Equipment Basis

System	Quantity	Description
Main transformers	3	Hitachi Energy 138/13.8 kV, 100/133/167 MVA OA/FA/FA
Main switchgear	1 lineup	Siemens NXAIR M 15 kV-class metal-clad switchgear
Data halls	6	Nominal 30 MW gross hall blocks; 20 MW critical IT per hall
Standby generators	60	Rolls-Royce mtu 16V4000 DS2500 at 2.5 MW each, 13.8 kV
UPS modules	216	Vertiv Liebert EXL S1 1,200 kW modules
Capacitor banks	6	6 Mvar switched steps on the 13.8 kV campus system